The §4 gate bound: the objects, the counterexamples, and exactly why each proof route fails — worked computations

(plus: why the target itself was FALSE, and the cube-shell bypass)

(secretary worked-reference; all values machine-computed from the faithful model)

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CRITICAL UPDATE (2026-06-16) — the central inequality of this note is FALSE.

The boxed target $\|\mathrm{scd}\, p\, k\| \le \mathrm{ctx\_bound}(\mathrm{drain\_ctxs} p)\, k$ (§1) is *not* true in-regime. Its "0 violations over $> 10^5$ cases" was a **sampling artifact**: the true violation rate is ≈1/1.2M, below what those gates sampled. Verified counterexample (S-fixed, normal form, depth≥ 5):

$$p = \mathbf{1} + \big(\,[\,\mathbf{1} + (\mathbf{1} + a)\cdot a + (c + \mathbf{1})\cdot(c + \mathbf{1})\,]\cdot c^*\big), \quad k = c^* \;\Rightarrow\; \|\mathrm{scd}\| = 47 \;>\; 46 = \mathrm{ctx\_bound}.$$

The *looser* fallback $\|\mathrm{scd}\, p\, k\| \le \mathit{drain\_child\_budget}\, p\, k$ — the predicate `child_ok`, which is what the gate actually needs — is **also false** (CE 99 > 96); its "0/1.2M" was the same artifact. So every route in §5 failed because **the target itself is false**, not for want of cleverness; the worked CEs are the local symptoms of that global fact. **Resolution** (§7): the gate needs *neither* per-step budget — it reduces, through already-green lemmas, to a *cube-shell* invariant that *is* true, leaving one star-case lemma.

Notation. Characters a, b, c; r^* = star; $r \cdot s$ = sequence; $r + s$ = alternation; $\mathbf{1}$ = the empty-string regex. $|\cdot|$ is regex size; $\|X\| = \sum_{x\in X} |x|$ over the *deduplicated* set X. All numbers below are computed by the validated Python model (`scratch_ce_trace_for_doc.py`).

1 The inequality these routes tried (now known FALSE)

The D law, the static cubic front, and the gate wrapper are all green. The effort reduced the §1 gate to the single numeric inequality

$\boxed{\|\mathrm{scd}\ p\ k\| \;\le\; \mathrm{ctx\_bound}(\mathrm{drain\_ctxs}\ p)\ k}$ (FALSE in-regime — sampling artifact; see the update box above).

This was *believed* true (0 violations over $\sim 10^4$–10^5 sampled depth-≥ 5 cases), and every route in §5 is an attempt to prove it. It is in fact false (rate ≈1/1.2M). The worked counterexamples below remain exactly correct and *instructive*: they are the local symptoms (the $a^* \cdot a^*$ collapse heterogeneity) of the global fact that the target is unprovable — and the *same* symptoms make the looser `drain_child_budget` target fail too. Read §2–§5 as "why no per-row/per-slot accounting can work," then §7 for the route that does.

2 The objects (concrete)

S (`rsimpStrong_raw`). The strong normaliser. Crucially it *collapses* $a^* \cdot a^* \to a^*$ (a guarded rule, only when the two starred bodies are syntactically equal).

$\diamond_4$ (`rsimp4_SEQ_atom`). The *weak* sequence plug: $h \diamond_4 k$ appends k after head h (reassociating), but **never** collapses $a^* \cdot a^*$. (`rsimp7` is the same but *does* collapse, at the top head only.)

drain_ctxs p. A *list* of slots $(\mathit{head}_i, \mathit{cost}_i)$, one per "drain weight" unit of p (one per character, one per star), built structurally. $\mathit{slot\_cost}(p, k, i) = \mathit{cost}_i + (1 + |k|)$ and $\mathrm{ctx\_bound}(\mathrm{drain\_ctxs} p)\, k = \sum_i \mathit{slot\_cost}(p, k, i)$.

$\boldsymbol{\mathit{weak\_slot\_rows}}(p, k, i) = \mathrm{rows}(\,\boldsymbol{\mathit{head}}_i \diamond_4 k\,) \setminus \mathrm{rows}(k)$**.** The *uncollapsed* opening of slot i. wcd p $k = \bigcup_i \mathit{weak\_slot\_rows}(p, k, i)$. Green fact: $\|\mathrm{wcd}\ p\ k\| \le \mathrm{ctx\_bound}(\mathrm{drain\_ctxs} p)\, k$.

scd p k = SOL($\boldsymbol{\mathit{nseq}}\, p\ k$) \ SOL(S$k$)**.** The **actual** object: the opened rows after the *strong*, S-normalised step. *These rows are collapsed.*

The whole tension in one line: the weak rows use $\diamond_4$ (uncollapsed), the strong rows use S (collapsed). So a strong row and "its" weak row are often *different rows* of *different sizes*, even though the budget covers them.

3 Three counterexamples, fully computed

All three use $k = a^*$, and all are in-regime ($\mathsf{S}p = p$, $\mathsf{S}k = k$, normal form). For each p the slot ledger drain_ctxs$p$ and ctx_bound are shown, then scd and wcd.

CE-A: $p = b \cdot a^*$ — the collapse, in isolation

$$\mathrm{scd}(b \cdot a^*,\, a^*) = \{\underbrace{b \cdot a^*}_{||=4}\}, \qquad \|\mathrm{scd}\| = 4.$$

Slots of drain_ctxs$(b \cdot a^*)$ (head, cost, slot_cost, weak row):

$$\begin{array}{llll} \mathrm{slot}_0: & b \cdot a^* \ (4) & \mathit{slot\_cost} = 7 & \mathrm{weak} = b \cdot (a^* \cdot a^*) \ \ || = 7 \\ \mathrm{slot}_1: & a^* \ (2) & \mathit{slot\_cost} = 5 & \mathrm{weak} = a^* \cdot a^* \ \ || = 5 \\ \mathrm{slot}_2: & a \cdot a^* \ (4) & \mathit{slot\_cost} = 7 & \mathrm{weak} = a \cdot (a^* \cdot a^*) \ \ || = 7 \end{array}$$

ctx_bound $= 7 + 5 + 7 = 19$, and wcd $= \{b \cdot (a^* \cdot a^*),\ a^* \cdot a^*,\ a \cdot (a^* \cdot a^*)\}$, $\|\mathrm{wcd}\| = 19$. **Key:** the one strong row $b \cdot a^*$ is the S-*collapse* of the slot_0 weak row:

$$\mathsf{S}\big(b \cdot (a^* \cdot a^*)\big) = b \cdot a^*, \qquad |b \cdot a^*| = 4 \ \le\ 7 = |b \cdot (a^* \cdot a^*)|.$$

So $b \cdot a^* \notin$ wcd (set-containment into the weak cover *fails*), but it is the collapse of a weak row, and collapse *shrinks*. Bound holds: $4 \le 19$.

CE-B: $p = b \cdot a^* + \mathbf{1}$ — the RALTS wrap *retains* the uncollapsed row

Same slots/ctx_bound = 19 as CE-A (the $\mathbf{1}$ adds no slot). But now

$$\mathrm{scd}(b \cdot a^* + \mathbf{1},\, a^*) = \{\underbrace{b \cdot (a^* \cdot a^*)}_{||=7}\}, \qquad \|\mathrm{scd}\| = 7.$$

Same head $b \cdot a^*$, opposite outcome: wrapped under $+$, the opener exposes the *uncollapsed* $b \cdot (a^* \cdot a^*)$ and the strong step does *not* collapse it back. And S runs the *wrong way* for any witness:

$$\mathsf{S}\big(b \cdot (a^* \cdot a^*)\big) = b \cdot a^* \ \neq\ b \cdot (a^* \cdot a^*).$$

So *no* weak row y has $\mathsf{S}(y) =$ the strong row (every S-image is the *shorter* $b \cdot a^*$). The strong row here *is* the slot_0 weak row itself ($x = y$). Bound holds: $7 \le 19$.

CE-Hall: $p = a + (b^* \cdot a^*)$ — both versions at once

$$\mathrm{scd}(a + b^* \cdot a^*,\, a^*) = \{\underbrace{a \cdot a^*}_{4},\ \underbrace{b^* \cdot a^*}_{5},\ \underbrace{b^* \cdot (a^* \cdot a^*)}_{8}\}, \quad \|\mathrm{scd}\| = 17, \quad \mathrm{ctx\_bound} = 34.$$

The slot for head $b^* \cdot a^*$ is slot_1 (cost 5, slot_cost 8, weak row $b^* \cdot (a^* \cdot a^*)$, size 8). Now:

$$b^* \cdot (a^* \cdot a^*) \text{ is the slot}_1 \text{ weak row } (x{=}y), \qquad \mathsf{S}\big(b^* \cdot (a^* \cdot a^*)\big) = b^* \cdot a^*.$$

So *both* strong rows $b^* \cdot a^*$ (collapsed, 5) and $b^* \cdot (a^* \cdot a^*)$ (uncollapsed, 8) have their *only* origin at slot_1, whose single budget 8 cannot pay both ($5 + 8 = 13 > 8$). Bound still holds ($17 \le 34$): the *total* budget covers them because other slots are underused.

4 Each failed route: left side, right side, value on the CE

R1. Per-child set-containment (`master_cover`, the original C-DRAIN). Tried: $\mathrm{scd}(\sum_q q)\, k \subseteq \bigcup_q \mathrm{scd}\ q\ k$ (parent rows are union of child rows). *Intuition:* a $\sum$ should just be the union of its parts. *Killed by CE-B:* LHS $= \{b \cdot (a^* \cdot a^*)\}$ (parent, uncollapsed), but the child $b \cdot a^*$ alone gives $\mathrm{scd}(b \cdot a^*, a^*) = \{b \cdot a^*\}$ (CE-A, collapsed) and $\mathbf{1}$ gives $\{\}$, so RHS $= \{b \cdot a^*\}$. The parent's uncollapsed row $b \cdot (a^* \cdot a^*) \notin \{b \cdot a^*\}$. The $+$-wrap blocks the collapse the child performs. Fails.

R2. Total-size domination. Tried: $\|\mathrm{scd}\ p\ k\| \ \le\ \|\mathrm{wcd}\ p\ k\|$ (then the green $\|\mathrm{wcd}\| \le$ ctx_bound finishes). *Intuition:* strong rows are "smaller" than the weak cover, so their total is too. *False:* S can *grow* a row's opened ledger, so the strong total can exceed the weak total. Machine CE: $p = (\,\mathbf{1} + b \cdot (a + \mathbf{1}) + (\mathbf{1} + a) \cdot a^*\,) \cdot a^*$, $k = c$: $\|\mathrm{scd}\| = 36 \ >\ 29 = \|\mathrm{wcd}\|$ (while ctx_bound = 51 still covers 36). So the weak *set* does not dominate; only the *ledger* over-covers.

R3. Injective slot-origin via `collapses_to` (verdict6). Tried: an injection $x \mapsto$ (slot i_x, witness y_x) with $\mathsf{S}(y_x) = \{x\}$ and $|x| \le |y_x|$. *Intuition:* each strong row is the collapse of a paid weak slot row. *Killed by CE-B:* the strong row $x = b \cdot (a^* \cdot a^*)$ is *uncollapsed*; no weak y has $\mathsf{S}(y) = x$ (S only *produces* the shorter $b \cdot a^*$). The relation points the wrong way: x is the long predecessor, not the collapse. (97/40849 failures.)

R4. Two-route injective charge (secretary salvage). Tried: same injection but $x = y_x \vee x \in \mathrm{rows}(\mathsf{S}y_x)$ (allow "*x is* the weak row"). *Intuition:* cover both the uncollapsed ($x{=}y$) and collapsed ($x = \mathsf{S}y$) cases. *Killed by CE-Hall:* $b^* \cdot a^*$ (via $x \in \mathsf{S}y$) and $b^* \cdot (a^* \cdot a^*)$ (via $x{=}y$) are *both* edged only to slot$_1$. Two rows, one slot $\Rightarrow$ no injection (a Hall collision), even though slot$_1$'s cost would suffice for either alone. (24/52851; cuts R3 by $4\times$ but cannot reach 0.)

R5. Pure numeric induction, RALTS step. Tried: $\|\mathrm{scd}(\sum_q q)\,k\| \;\le\; \sum_q \|\mathrm{scd}\; q\; k\|$ (deduped-union size bounded by the sum of children). *Killed by CE-B:* LHS $= \|\{b \cdot (a^* \cdot a^*)\}\| = 7$, RHS $= \|\{b \cdot a^*\}\| + 0 = 4$; $7 > 4$. The parent's single uncollapsed row is *bigger* than the whole child sum. (RALTS has zero constructor slack, so there is no room for the extra.)

R6. Any slot-derived carrier (the impossibility square). For *any* carrier W' built from the slot rows: plain weak $W' = \mathrm{wcd}$ has $\|\mathrm{scd}\| > \|W'\|$ sometimes (R2); S-applied carriers shrink *below* scd (bridge fails worse); wcd$\,\cup\,$scd pushes the *ceiling* above ctx_bound (CE-A-like: $23 > 19$). No carrier sits between $\|\mathrm{scd}\|$ and ctx_bound. The whole carrier/membership class is dead.

5 The one obstruction, stated once

scd $p\;k$ is a *heterogeneous* set: a $\sum$/star wrap can make it *retain the uncollapsed long row* $b \cdot (a^* \cdot a^*)$ (CE-B), while a bare sequence makes it *keep the collapsed short row* $b \cdot a^*$ (CE-A), and a mix keeps *both* (CE-Hall). The weak cover only ever holds the uncollapsed forms. Therefore:

- set-containment fails (collapsed strong rows are not weak members);
- S-witnessing fails (uncollapsed strong rows are not S-images);
- any row$\to$slot *injection* fails (collapsed + uncollapsed siblings collide on one slot);
- yet the *total* ledger ctx_bound always covers $\|\mathrm{scd}\|$ (other slots are underused).

Consequence (the search-narrowing result): the proof cannot be a per-row or per-slot *set/membership* statement. It must be a *total-budget* argument that does not match rows to slots.

6 Two further routes (node-to-atom), also refuted

verdict7 ("budgeted collapse trace") charged at the level of *syntax-node occurrences $\to$ cost tokens* (atoms of ctx_bound), not rows $\to$ slots, hoping the short row $b^* \cdot a^*$ and the long row $b^* \cdot (a^* \cdot a^*)$ draw *disjoint* atoms. Modelled as a max-flow it is feasible in 40851/40852 cases but fails one: parallel star-children sharing a continuation over-subscribe the single source-star slot — the long predecessor *contains a full copy* of the short row's body, so the atoms are not disjoint. **verdict8** (occurrence-keyed traced budget) keys events by producer-path; the key is honest but powerless (it only re-routes a fixed total, never shrinks it), and the unowned doubled-star rows cannot be keyed. Both die on the same wall — and, as the update box records, on the deeper fact that the target inequality is false.

7 The resolution (2026-06-16): drop the per-step budget; the cube-shell bypass

Both per-step budgets are false in-regime: the tight ctx_bound (CE $47 > 46$) and the looser *drain_child_budget* $=$ *open_pot* p + *zw2p* $\cdot\,(1{+}|k|)$ that `child_ok` needs (CE $\|\mathrm{scd}\| = 99 > 96$). Their "0-violation" validations were sampling artifacts: the killer family — nested $\cdot\big(+[\mathit{pre}, \mathbf{1}],\ \ldots \cdot (\mathrm{atom}, \mathrm{star}^*)\big)$ opened at $k = \mathrm{star}^*$, which re-doubles $\mathrm{star}^* \to \mathrm{star}^* \cdot \mathrm{star}^*$ at every frame — is never built by uniform random generation. The deficit grows *quadratically* with chain depth, so no constant, multiplier, or minimal companion rescues a $\sim$quadratic budget.

But the gate never needed a per-step budget. It already has a second, fully green route in the theory: the wrapper hypothesis $\|\text{one strong step of the front}\| \le 2(|r|+3)^3$ follows from a *per-row* bound $\|\mathrm{SOL}(r)\| \le \mathit{pot}(r, \mathbf{1})$ (green, structural induction) plus the *cube-shell invariant*

$$\boxed{\mathit{pot}(r,k) \;\le\; (|r|+|k|)^3 - (|k|)^3} \qquad \text{(machine-validated TRUE, 0/90k+ incl. the killers above).}$$

This succeeds where the budgets failed because it bounds the per-row *opening universe* by a *cubic* telescoping shell (a difference of cubes, which composes *additively*), not the collapsed+uncollapsed shadow rows by a quadratic budget. Leaves ($\mathbf{1}$, char) and the $+/\cdot$ steps are green; the lone open step is **RSTAR**, which decomposes (validated) into a routine HEAD term plus the saturation

$$\text{SAT:}\quad \mathit{pot}\big(r,\; r^*\cdot k\big) \;\le\; (|r|+|k|)^3 - (|k|)^3$$

— opening the star body r against its *own* star costs only the body's k-shell (the r^* prefix is absorbed for free). SAT is true with margin (ratio ≤ 0.81) and is the single remaining piece; full plan in `RSTAR_CUBE_SHELL_SKETCH.md`.